

Rate

* Rate: A measure of the change that happens in the reaction per unit time.

2 ways to find the rate of a rxn:

- 1- Measure the amount of reactants used up per unit time.
- 2- Measure the amount of products produced per unit time.

Ways of measurement:

1- The change of mass over time

2- The change in the volume of gas given over time
↳ gas is measured by a gas syringe or upside-down measuring cylinder

$$\text{Rate of rxn} = \frac{\text{amount of reactant used / amount of product produced}}{\text{time}}$$

3- Measuring the pH change. / Temperature changed. / Change of pressure. / Electrical conductivity.

To calculate: (Gas)

$$* \text{Rate} = \frac{\text{Volume}}{\text{time}} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$* \text{Average rate} = \frac{\text{Total volume}}{\text{total time.}}$$

→ The rate of reaction is greatest at the start of the rxn, but it decreases as the reaction proceeds

→ At the start of the rxn, the curve of the line is the steepest. Faster the rxn → Steeper the curve

→ When no more products are produced, the volume/mass won't change. So curve goes flat.

Reaction over → Flat curve.

→ The initial rate of a rxn is the greatest (steepest curve) because the concentration of the reactants is the highest, so more frequent collisions.

→ The reaction's curve becomes less steep because the concentration of the reactants decreases, so less frequent collisions.

* Chemical reactions only happen if reactant particles have enough energy in the correct orientation. The more frequently particles collide, and the greater the proportion of collisions with enough energy, (E_a), the greater the rate.

* Particles must have minimum amount of energy to collide with other particles and react.

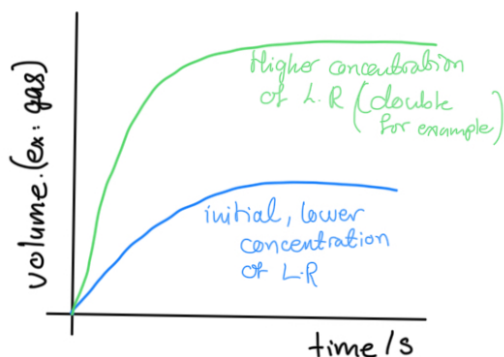
* The minimum energy required is activation energy. (E_a)

① If the concentration of a reactant is increased:

- More reactant particles in a given volume
- So, there is a greater chance of the particles colliding successfully per unit time.
- The rate of rxn increases.

Effect of limiting and excess reactants on the rate of reaction and volume of products:

1- Using same volumes, but different concentrations of limiting reactant.



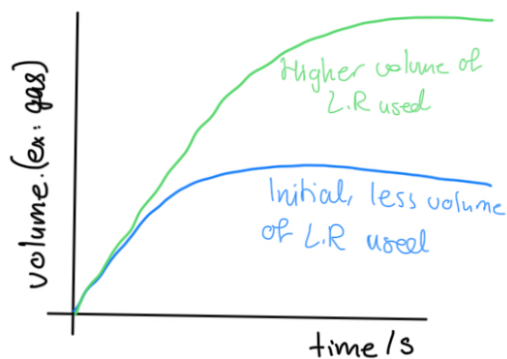
Conclusion:

→ Increasing the concentration of L.R will increase the rate, curve is steeper.

→ Increasing the concentration at same volume will give more amount of product

→ Doubling the concentration will double the amount of products produced.

2- Using same concentration but different volumes of L.R

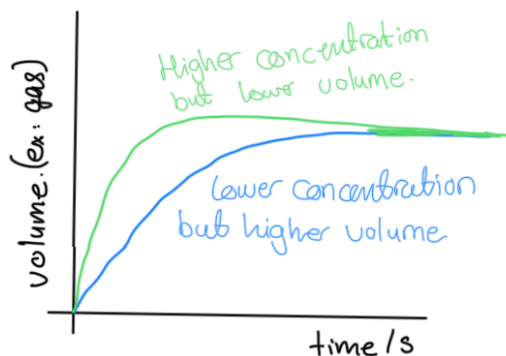


Conclusion:

→ Same steepness initially because same concentration is used.

→ Increased volume of product is produced.

3- Different volumes & different concentrations of L.R.



Conclusion:

→ Initial rate of second reaction becomes steeper because more concentration is used.

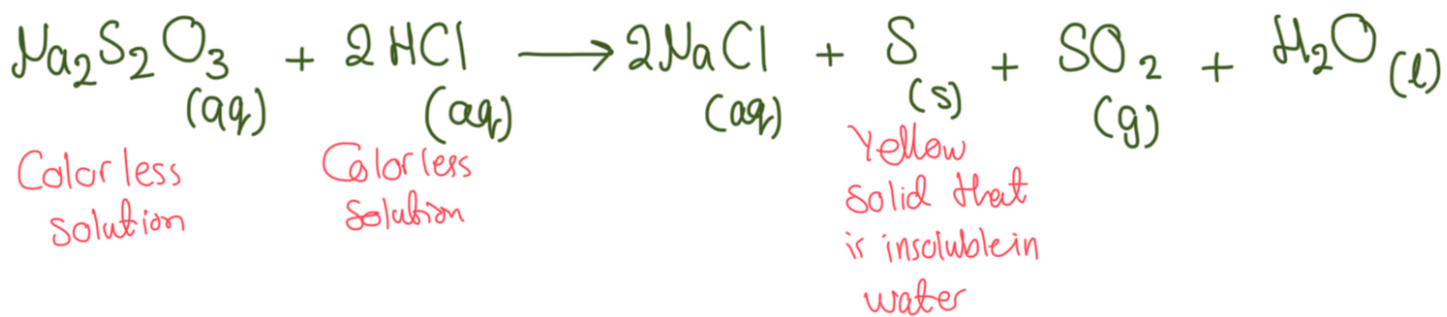
→ Same volume of product is produced because:

* You will either half the volume and double the concentration of L.R

OR

* Double the volume and half the concentration of L.R

Sodium + Hydrochloric → Sodium + Sulfur + Sulfur + Water
thiosulfate acid chloride dioxide



A yellow precipitate is formed from the reaction of sodium thiosulfate and hydrochloric acid. The rate of this reaction can be followed by recording the time taken for a given amount of sulfur to be precipitated.

Reaction:

- 1- Place a conical flask containing the reaction mixture onto a cross drawn on a piece of paper.
- 2- As the precipitate of sulfur forms, the cross is obscured and finally disappears from the view.
- 3- The time taken for this to occur is a measure of the rate.
- 4- Several experiments on this reaction should be made to obtain sufficient information.
- 5- These experiments will contain different concentrations of sodium thiosulfate or hydrochloric acid.
- 6- Since we're testing the effect of concentration, the overall volume of the mixture should be the same for all experiments.

② In a reaction only involving gases, an increase in the overall pressure will increase the rate.

→ Increasing the pressure of reacting gases

→ More gas molecules in a given volume

→ More collisions per unit time, as the pressure results in the gas particles being pushed closer together.

→ Rate of reaction increases as particles collide more and react faster.

* Increasing the pressure of a gas (by squeezing it in a smaller volume, giving it less volume), will increase its concentration.

Concentration by mass $\equiv \frac{\text{mass}}{\text{volume}} \rightarrow \text{g/dm}^3$

③ Surface area by changing particle size:

What will happen if a solid reactant is broken into smaller/powdered pieces?

1- Its surface area will be increased.

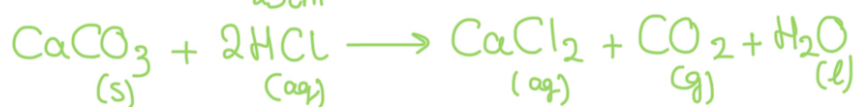
2- So more particles are exposed to the other reactant.

3- So there is a greater chance of the particles colliding → more frequent collisions.

$$\text{Surface area} \propto \frac{1}{\text{size}}$$

4- Rate of reaction increases.

Example: 0.05M
25cm³



Exp. 1: Large piece: 1g

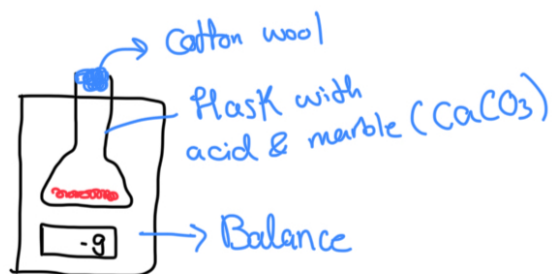
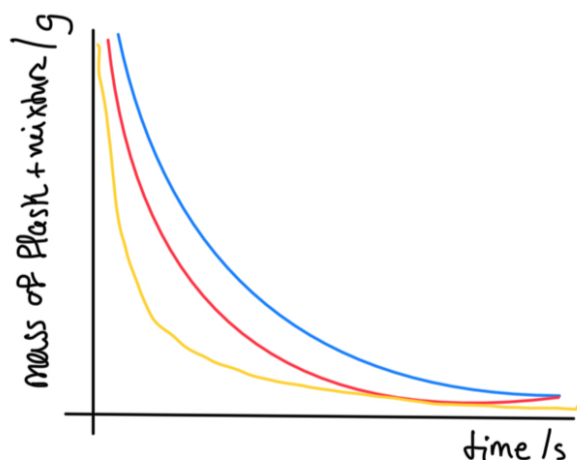
Exp. 2: 1g: Smaller pieces

Exp. 3: 1g: Crushed/powdered

Conclusion:

→ Crushed pieces had the fastest rate → steepest curve

→ all lost the same mass because same mass of CaCO₃ is used (1g)



⌚ → stopwatch.

* Cotton wool stops any liquids from splashing out.
And allows CO₂ to escape, so we can measure mass loss.

* Mass of CaCO₃ should stay the same.

* Volume & concentration of HCl should stay the same

* Temperature should stay same

* Same time intervals

When you increase the temperature in a reaction:

→ Reactant particles gain more kinetic energy.

→ So they move more quickly.

→ More particles have energy that is equal to the activation energy or greater.

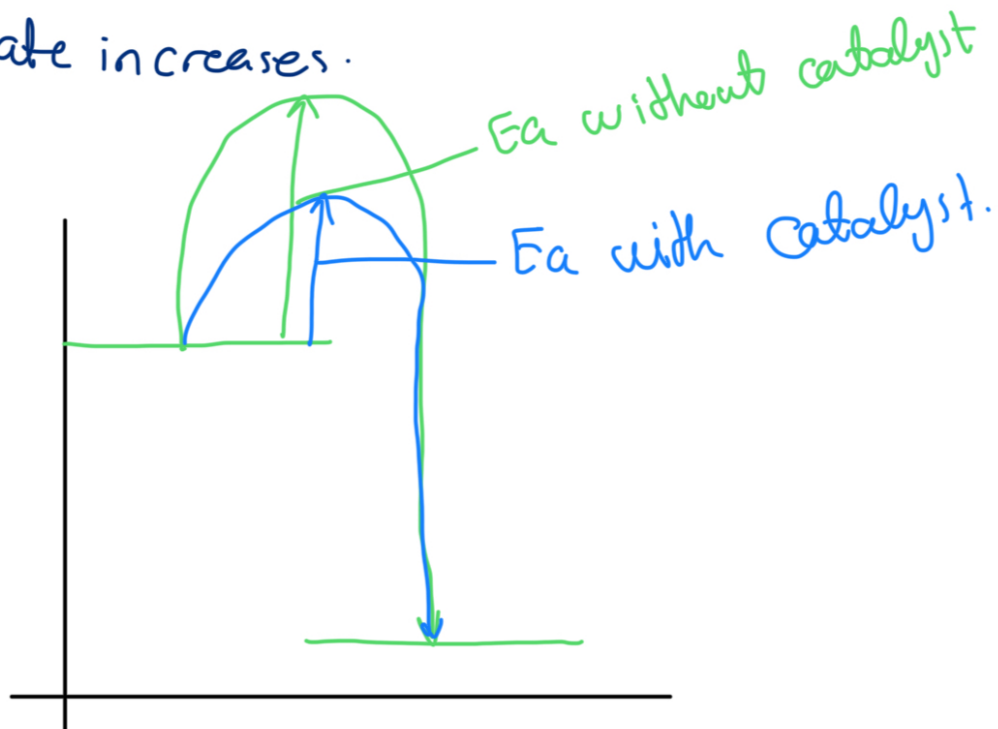
→ The particles will collide more frequently, and so more of the collisions result in reactions.

→ The rate of reaction increases.

Catalyst: A substance that increases the rate of a chemical reaction without getting used up in the reaction.

→ Catalyst increase the rate of reaction by lowering the activation energy needed by providing an alternative pathway, so a greater proportion of the molecules in the reaction will have sufficient energy to react

→ Using a catalyst will result in more collisions, so rate increases.



Hydrogen peroxide, decomposes to form water and oxygen.



* The reaction is too slow at room temperature.

So, manganese (IV) oxide: MnO_2 , a black solid, that acts as a catalyst to speed up the rate of the reaction.

→ The black powder (MnO_2) doesn't disappear through the reaction, and the mass remains the same at the end of the reaction.

mass of catalyst initially = mass of catalyst finally.

→ Manganese (IV) oxide can be filtered, rinsed and dried, to be used again

* The rate of reaction could be increased further if you increase the surface area of the catalyst.

* If dirt or impurities are present on the surface of the catalyst, the catalyst will not react as efficiently.

- * Small amount of catalyst will produce a large amount of chemical change.
- * Catalysts remain unchanged chemically.
- * Catalysts are specific to a particular chemical reaction.

Evidence for providing a larger surface area for the reactant to react faster:

→ Doubling the amount of the catalyst doubles the rate of the reaction.

→ Using finely powdered catalyst also increases the rate of the reaction.

Examples of catalysts:

1- Solid catalysts: transition elements or their oxides

2- Biological catalysts: Enzymes are proteins that are very specific as they control one reaction. Temperature sensitive, they can become denatured (inactive) above 45°C .

They are also sensitive to pH, most enzymes work best at neutral pH of 7

Pollutant gases released from car exhausts:

* Carbon monoxide: **Toxic**: Formed from the incomplete combustion of hydrocarbons in the fuel

* Nitrogen (II) oxide (NO): **Acid rain & respiratory products**:
Formed by the reaction of nitrogen gas with oxygen gas from the air at high temperature in the car engines and hot furnaces.



* Unburnt hydrocarbons.



To remove these gases: We use a catalytic converter.
→ It converts these harmful gases into less harmful products, such as carbon dioxide & water.

The converter has two compartments:

A: Harmful compounds are reduced. NO $\leftarrow$ $\begin{matrix} \text{nitrogen: } \text{N}_2 \\ \text{oxygen: } \text{O}_2 \end{matrix}$

B: Harmful compounds are oxidized, using the oxygen from A.

Carbon monoxide + oxygen → Carbon dioxide.

Unburnt hydrocarbons + oxygen → Carbon dioxide + Water.

The unharmed products flow out of the exhaust.

* Converter contains a thin coating of rhodium & platinum catalysts, which provide a large surface area for reaction.

* Converter can only be used with unleaded petrol, because lead poisons the catalyst.